

The Geography of the Attack

Dataset Description and Exploratory Analysis

Data Visualization — Technical Report (Deliverable 2)

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This report describes the dataset used for The Geography of the Attack and evaluates whether it is suitable for the proposed visualizations. The aim is to analyze attacking performance by attack zone and attack tempo, identifying which situations are most effective for Brescia and Talmassons, the two teams promoted from the Italian Women's Serie A2 in the 2025/26 season.

When comparing the two teams, grouped bar charts are used to show the differences between the corresponding values for each attack zone. Ordered bar charts rank attacking situations by efficiency, while seaborn jointplots illustrate the relationship between two variables together with their distributions. Finally, separate heatmaps are used for each team instead of a single combined heatmap, making it easier to compare their attacking patterns over the course of the promotion season.

1. Dataset Origin and Structure

The dataset consists of 63 DataVolley scouting files (.dvw) from the 2025/26 Italian Women's Serie A2 season. It includes 36 matches played by Valsabbina Millennium Brescia and 29 matches played by Talmassons, with the two teams facing each other twice during the season. Overall, the dataset covers the regular season, the playoff pool, the promotion playoffs, and the Coppa Italia.

Each .dvw file is divided into several sections, including match information, team rosters, attack combinations, and scouting data: [3MATCH], [3TEAMS], [3PLAYERS-H], [3ATTACKCOMBINATION], and [3SCOUT]. The [3SCOUT] section contains the rally-by-rally actions and is the main source of information used in this project. Across the 63 matches, the dataset contains 69,943 actions. Each action is stored using the same 27-field structure, with fields separated by semicolons (;), providing a consistent format for the analysis.

Each action is decoded to identify the team, player, skill type (serve, reception, set, attack, block, dig, or freeball), and the corresponding DataVolley evaluation code, which ranges from a perfect action to an error. For attack actions, the attack combination code is matched with the corresponding entry in the [3ATTACKCOMBINATION] section of the same match to identify the attack zone, side, and tempo. All 16,301 attack actions were successfully matched to a valid attack combination.

Because .dvw files are not accompanied by public documentation, interpreting the compact action codes required an initial decoding phase. This process was supported by the team's scout analyst, who explained the scouting conventions used during data collection. Table 1 summarizes the attack coding scheme adopted by DataVolley.

Field	Code	Meaning
zone	2	Zone 2 (opposite hitter / position 2)
zone	3	Zone 3 (middle hitter)
zone	4	Zone 4 (outside hitter / position 4)
zone	8	Zone 8 (pipe attack)
zone	9	Zone 9 (back-row attack from the diagonal)
tempo	Q	Quick attack
tempo	H	High ball
tempo	U	Super / medium tempo
tempo	M	Medium tempo
tempo	O	Other
tempo	N	No attack / rare case
tempo	T	Fast tempo
side	R	Right-side approach
side	L	Left-side approach
side	C	Middle approach

Table 1. DataVolley code definitions for attack zone, tempo, and approach side.

As an example, a single attack entry in the raw file is shown below:

*16AU#VS~45~H2;s;r;;;20.26.39;1;2;2;1;2074;;4;11;16;3;18;5;9;16;12;1;18;3;

The string before the first semicolon contains the action code, which is automatically decoded using the pydatavolley library. In this example, “*” identifies the home team, “16” is the player’s jersey number, “A” indicates the skill (attack), “U” the attack tempo, and “#” a point-scoring attack. The following two characters (“VS”) represent the attack combination code, which is matched to the corresponding entry in the DataVolley attack-combination table. In this case, “VS” corresponds to zone 4, right-side approach, and Super tempo.

2. Exploratory Data Analysis

Figure 1 shows the distribution of the recorded skills across the dataset. Since the objective is to compare the frequency of each skill, a simple bar chart is used. Attack is the most frequent skill, with 16,301 recorded actions, followed by Set (15,529), Serve (10,263), Dig (10,236), Reception (9,005), Block (7,029), and Freeball (1,580). As the aim of this project is to analyze attacking performance, the remainder of the analysis focuses on attack actions.

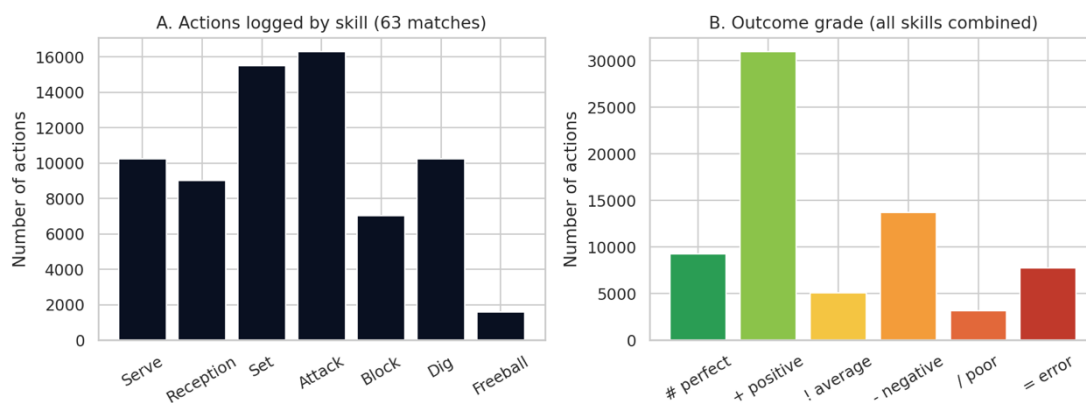


Figure 1. A) Number of logged actions by skill. B) Distribution of DataVolley evaluation codes across all skills.

Figure 2 compares the attack-zone distributions of the two promoted teams. A grouped bar chart is used to show, for each attack zone, the number of attacks recorded by Brescia and Talmassons. The two teams display very similar distributions in Zones 8 and 9, whereas the largest difference is observed in Zone 4, where Brescia records more attacks than Talmassons.

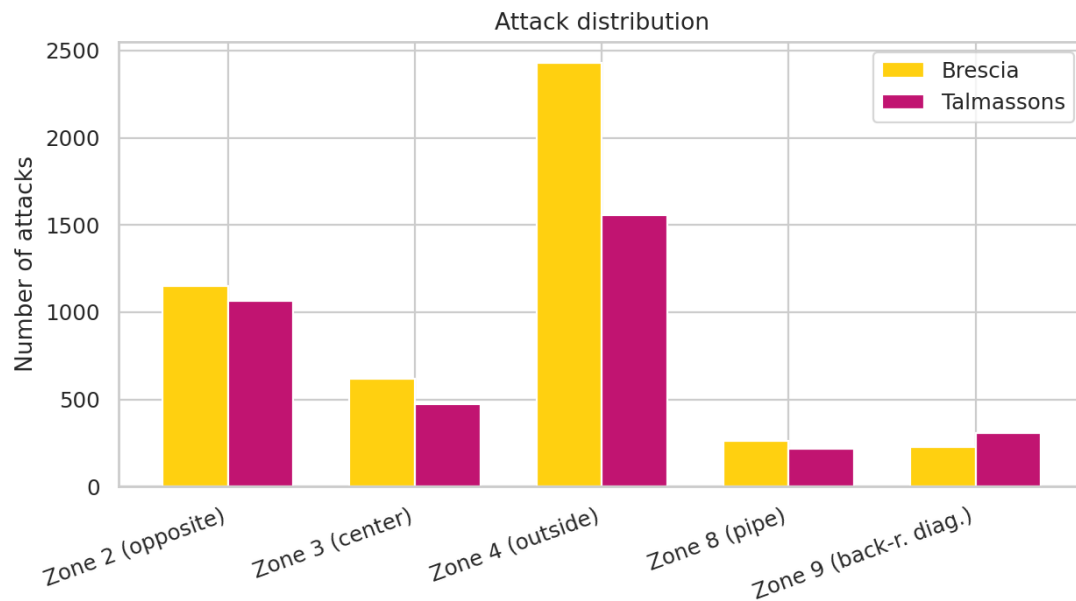


Figure 2. Attack distribution by court zone for Brescia and Talmassons.

For readers less familiar with volleyball, the five attack zones used in this report correspond to specific court positions rather than an arbitrary numbering system. Zones 4, 3, and 2 represent front-row attacks played close to the net (outside, middle, and opposite, respectively), while Zones 8 and 9 represent back-row attacks from behind the 3-meter line. These attacks generally take longer to develop but are often more difficult for the block to anticipate. Figure 3a illustrates the five attack zones considered in this analysis. In the original DataVolley scouting system, however, the court is divided into nine zones, as shown in Figure 3b.

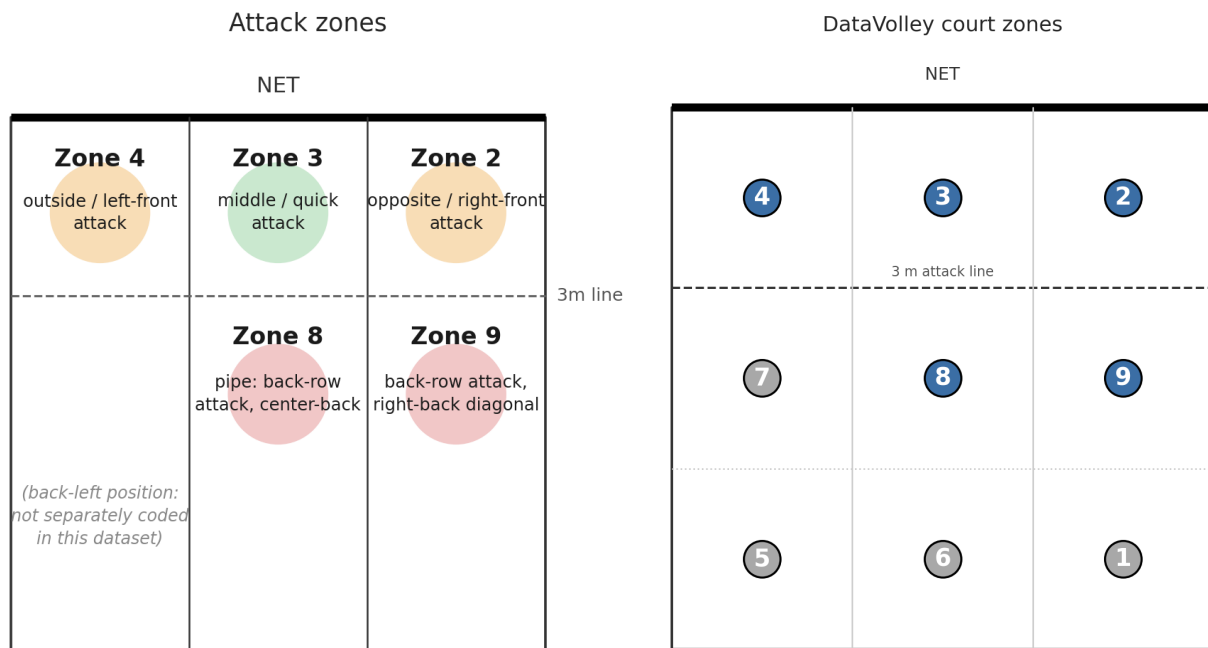


Figure 3. A) Schematic half-court showing the five attack zones used in this report. The back-left position is not separately coded in the scouting data. B) Court division used by DataVolley

Figure 4 ranks the five attack zones according to their net efficiency (kill rate – error rate), combining attacks from both teams. The zones are ordered from highest to lowest net efficiency to make the ranking easier to interpret. Zone 3 (middle attack) has the highest net efficiency (+0.43, $n = 1,089$), while Zone 8 (pipe attack) has the lowest (+0.24, $n = 479$). This result reflects the high effectiveness typically associated with quick middle attacks.

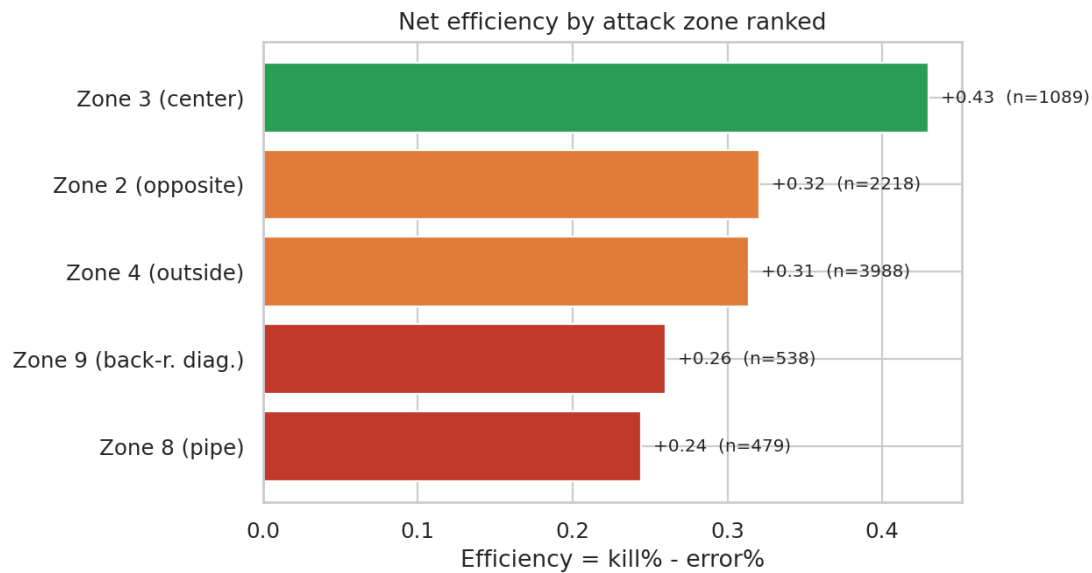


Figure 4. Net efficiency by attack zone for the two teams combined. Sample size is shown next to each bar.

Figure 5 compares the two main variables considered in this project: attack zone and attack tempo. Separate heatmaps are presented for Brescia and Talmassons using the same color scale, allowing a direct comparison between the two teams. Overall, the two heatmaps show a similar pattern. Both teams achieve their highest net efficiency on quick and medium-tempo attacks through the middle, while high-ball attacks to the back row are generally less effective. One difference can be observed in high-ball attacks to Zone 2, where Brescia records a lower net efficiency (0.20, $n = 320$) than Talmassons (0.34, $n = 256$). Whether these differences reflect real tactical differences or simply variation due to sample size is discussed in Section 3.2.

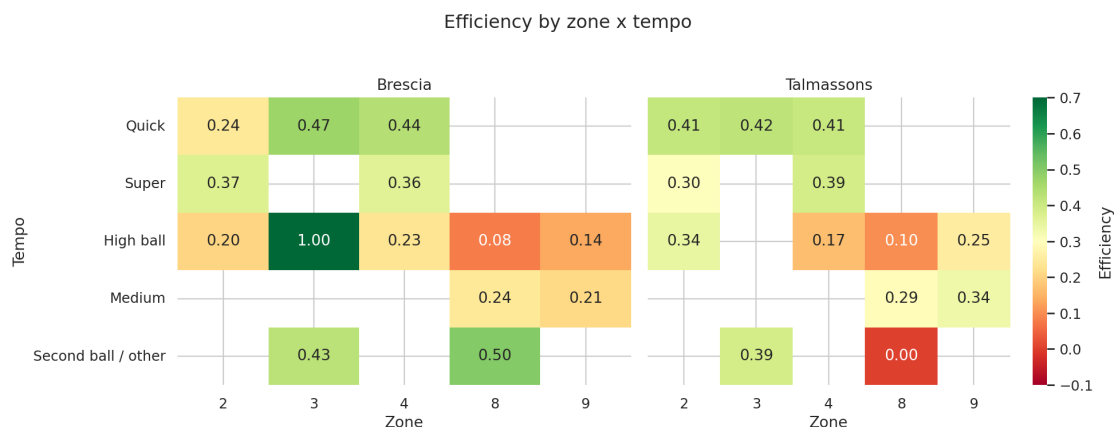


Figure 5. Attack efficiency by attack zone and tempo for Brescia and Talmassons.

Figure 6 complements Figure 5 by showing the number of attacks available for each zone $\times$ tempo combination. This makes it easier to distinguish between combinations based on many observations and those based on only a few. In the pooled dataset, 2 of the 14 observed zone $\times$ tempo combinations contain fewer than 30 attacks, while when the data are split by team, as in Figure 5, some of these sample sizes become even smaller.

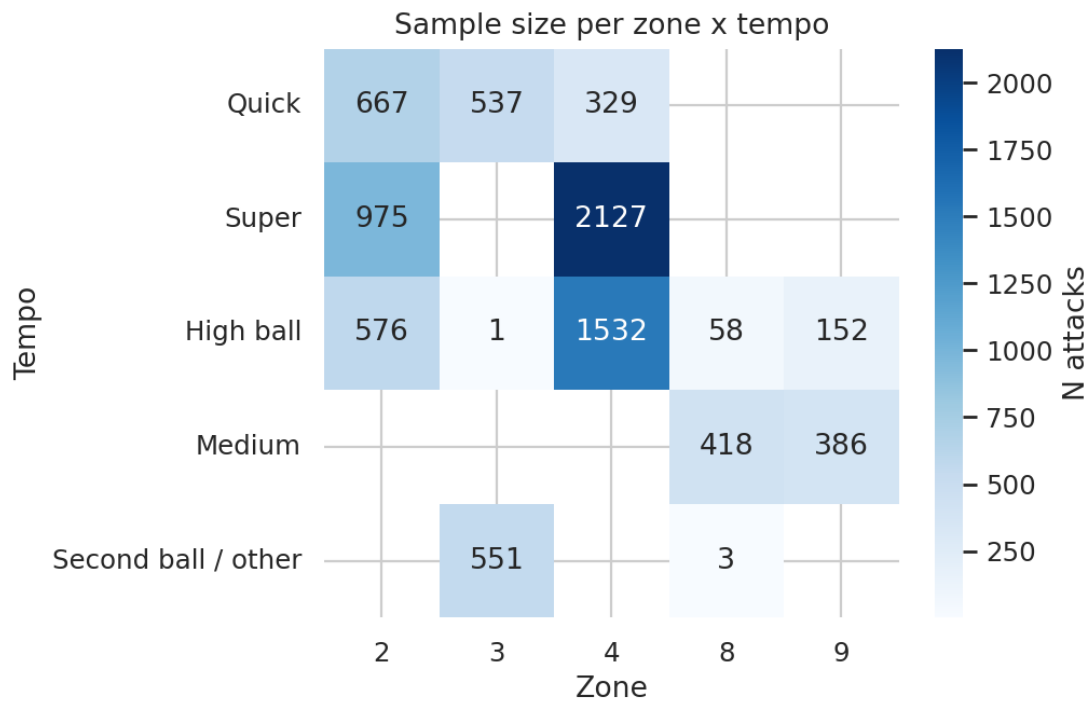


Figure 6. Number of attacks for each zone $\times$ tempo combination, with data from both teams pooled.

Figure 7 explores the relationship between the number of attacks in a match and overall attack efficiency. A seaborn jointplot is used to display both the relationship between the two variables and the distribution of each variable separately. Most matches contain between 100 and 150 attacks, as shown by the marginal distributions.

The relationship is clearly negative (correlation ≈ -0.66 across the 65 team–match observations included in the analysis): matches with more attacking actions tend to have lower attack efficiency. One possible explanation is that longer, more balanced matches include more rallies and transition attacks, reducing the probability of scoring from each individual attack. The distributions for Brescia and Talmassons are also very similar, suggesting that the two teams experienced comparable match profiles over the course of the season.

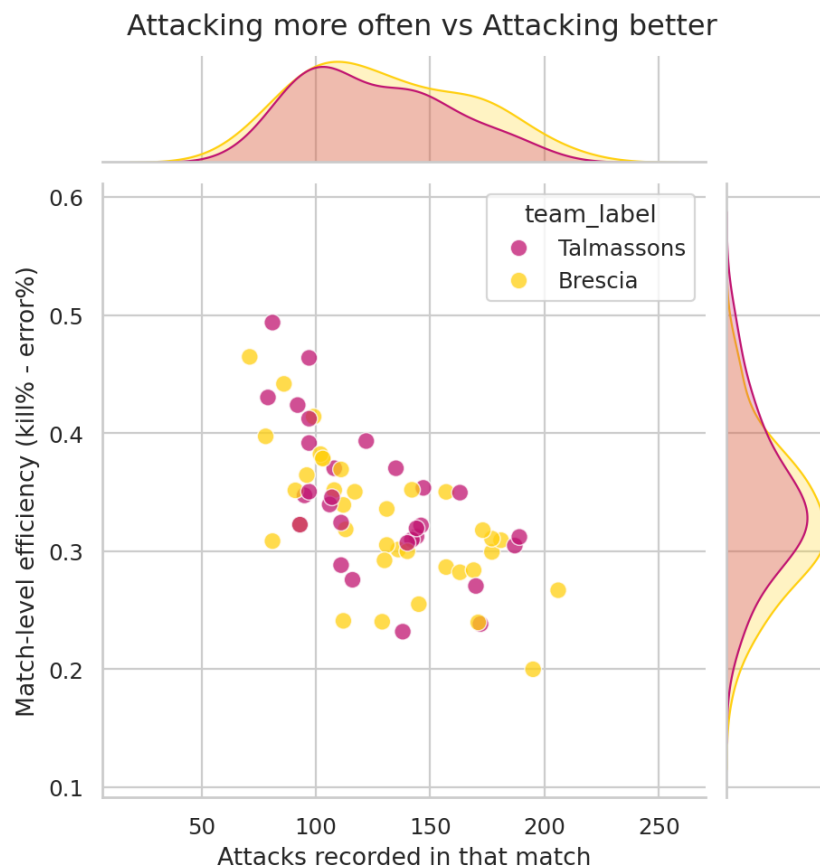


Figure 7. Match-level attack volume versus match-level attack efficiency, with marginal distributions coloured by team.

3. Assessment of Data Quality

3.1 What works well

- **All attack actions were decoded successfully.** Every attack was matched to a valid attack zone and tempo using the corresponding attack-combination table for each match, with no unmatched attack actions.
- **Complete player information.** All 69,943 recorded actions were linked to the corresponding player.
- **Consistent match counts.** The number of parsed matches (36 for Brescia and 29 for Talmassons, including the two matches between them) matches the values reported in the project proposal, confirming that all files were parsed correctly.
- **Consistent scouting profiles.** Figure 8 compares the distribution of DataVolley evaluation codes in the scouting files of Brescia and Talmassons using a grouped bar chart. The two distributions are almost identical (14.2% vs. 14.4% for the perfect grade and 45.0% for the positive grade in both teams), indicating that the scouting data were collected in a highly consistent way and can therefore be compared with confidence.

The similarity between the two datasets also makes this comparison meaningful. Although Talmassons achieved direct promotion while Brescia was promoted through the playoffs, the two teams finished the regular season only one point apart. This reduces the risk that the observed differences simply reflect a large difference in team strength and makes tactical explanations more plausible.

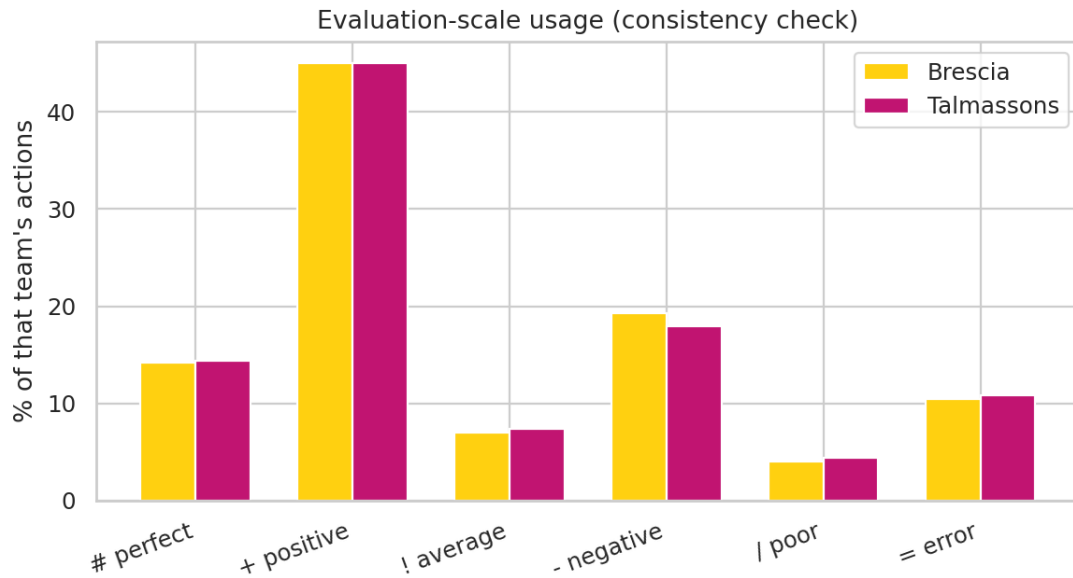


Figure 8. Distribution of DataVolley evaluation codes for Brescia and Talmassons shown with a grouped bar chart.

3.2 Limitations to carry into the visualization design

- **Home/away information.** The home/away field cannot be interpreted as the actual match venue because self-scouted DataVolley files always list the scouting team first. This variable would therefore need to be checked against the official match schedule before being used in the analysis.
- **Small sample sizes.** Some zone $\times$ tempo combinations contain only a few attacks. After splitting the data by team, these sample sizes become even smaller. For example, Zone 8 $\times$ second-ball attacks include only 2 observations for Brescia and 1 for Talmassons. These cases are not interpreted as meaningful results, unlike the difference observed for high-ball attacks to Zone 2, which is based on several hundred observations for each team. They are not removed from the dataset because there is no reliable way to distinguish a rare but valid attack from a possible scouting error. Instead, they are retained and interpreted with caution.
- **Number of opposing blockers.** The pydatavolley library extracts this variable from a fixed position in the DataVolley action code. The values from 0 to 3 behave as expected, with attack efficiency decreasing as the number of blockers increases (Figure 9). However, a fourth value also appears. Since an attack cannot face four front-row blockers, this value required further investigation before the variable could be used (Section 4). It appears consistently, although at a low frequency, across all 63 matches. Public documentation explains its meaning: it corresponds to the DataVolley category 'Hole block', meaning that the attack passes through a gap in

the block rather than facing a literal player count. For this reason, it is retained as a separate category rather than merged with “3” or removed from the analysis (Figure 10).

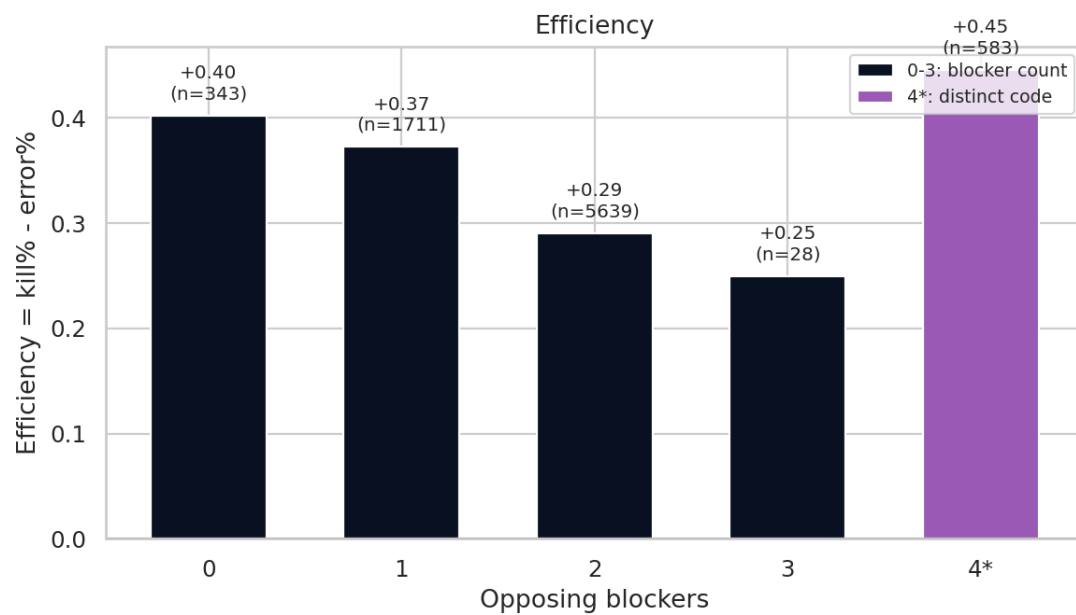


Figure 9. Efficiency by number of opposing blockers. Efficiency decreases as the number of blockers increases from 0 to 3. The fourth category (“4*”) is displayed separately.

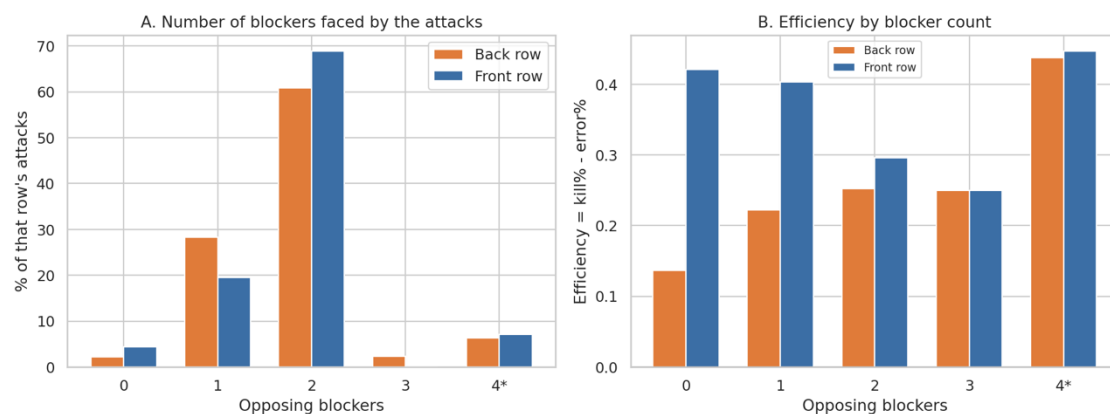


Figure 10. A) Distribution of the number of opposing blockers faced by front-row attacks (Zones 2, 3, and 4) and back-row attacks (Zones 8 and 9). B) Attack efficiency by blocker count for front-row and back-row attacks.

- **Subjective scouting:** DataVolley evaluations are assigned by human operators during the match, so small differences in coding are unavoidable and cannot be completely removed during data processing. Discussions with the team’s scout analyst were particularly valuable for understanding the scouting process and interpreting ambiguous cases. This made it easier to distinguish genuine patterns in the data from possible scouting noise.

4. Technical Correctness and Reproducibility

The analysis pipeline was implemented in Python. Action codes were decoded using the pydatavolley library (openvolley/pydatavolley on GitHub), with additional code written to read the match-specific [3ATTACKCOMBINATION] table for the structured tempo field and to derive the match phase from the file name. Data processing and visualizations were performed using pandas, Matplotlib, and Seaborn.

Before starting the analysis, several validation checks were carried out. These included verifying that every scouting row contained the expected 27 fields, confirming the total number of matches, checking that all attack actions were successfully matched to a valid attack zone and tempo, and validating the derived variables.

During this process, the num_blockers field required further investigation because it contained an additional value besides the expected blocker counts (0–3). Rather than assuming it was an error or merging it with the “3 blockers” category, it was examined in more detail.

The attack zone provided by pydatavolley (start_zone) was compared with the corresponding [3ATTACKCOMBINATION] table. The two sources matched for all 8,312 focal attacks, confirming that the action codes were decoded correctly. The frequency of the "4" category was then examined across the whole dataset. It was present in all 63 matches, with values ranging from 1.9% to 15% of attacks (mean $\approx$ 6.6%). In addition, 56% of these attacks were quick-tempo attacks, compared with 14% in the rest of the dataset. This distribution is consistent with the documented meaning of the value (Hole block), which describes an attack passing through a gap in the opposing block rather than a literal blocker count.

The original DataVolley files, the processed datasets, and all analysis scripts are stored together so that the complete workflow can be reproduced from the raw data.

5. Implications for the Visualization Deliverable

The analyses presented in this report show that the proposed visualizations can be built from the available data. The grouped bar charts, ranked bar chart, heatmaps, and jointplot all provide useful information when applied to the full dataset.

Attack zone and attack tempo were successfully extracted for every attack action. After investigating the unexplained fourth blocker category (Section 4), the number of opposing blockers was also considered suitable for analysis. The comparison between Brescia and Talmassons already reveals several common patterns as well as some differences between the two teams (Figures 2, 5, and 8).

The final Bokeh application will follow the same approach by presenting attack efficiency together with the corresponding sample size. Displaying both pieces of information allows users to distinguish between patterns supported by many observations and results based on only a few attacks.

6. References

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5. Bokeh Development Team. *Bokeh*: Python library for interactive visualization. <https://bokeh.org>
6. Slides of the course *Big Data Technologies and Visualization* [136069], Moodle.